

2020—2021 学年春《高等数学》(II) 考试试卷 (A) 卷

参考解答及评分标准

一、单选题 (共 5 小题, 每小题 2 分, 共 10 分)

1. A 2. B 3. B 4. D 5. C

二、填空题 (共 5 小题, 每小题 2 分, 共 10 分)

6. $(5, 3, 2)$ 7. -9900 8. $3x - 4y + z = 0$ 9. 4 10. -4π

三、解答题 (共 11 小题, 共 80 分)

11. (6 分) 【解】由平面的截距式方程, 可知由 $A(1, 0, 0)$, $B(0, 2, 0)$, $C(0, 0, 3)$ 所确定的平面方程为

$$\frac{x}{1} + \frac{y}{2} + \frac{z}{3} = 1.$$

将 D 点的坐标 $(-1, 2, t)$ 代入上面的方程可得 $t = 3$. (3 分)

易知 $\overrightarrow{AB} = (-1, 2, 0)$, $\overrightarrow{AC} = (-1, 0, 3)$, $\overrightarrow{CD} = (-1, 2, 0)$, 故四边形 $ABCD$ 为平行四边形, 从而其面积为

$$S_{ABCD} = |\overrightarrow{AB} \times \overrightarrow{AC}| = \left\| \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 2 & 0 \\ -1 & 0 & 3 \end{vmatrix} \right\| = |(6, 3, 2)| = 7. \quad (6 \text{ 分})$$

【另解】利用空间四点 A, B, C, D 共面的充要条件, 即三向量

$$\overrightarrow{AB} = (-1, 2, 0), \overrightarrow{AC} = (-1, 0, 3), \overrightarrow{AD} = (-2, 2, t)$$

共面, 或即

$$(\overrightarrow{AB} \times \overrightarrow{AC}) \cdot \overrightarrow{AD} = \begin{vmatrix} -1 & 2 & 0 \\ -1 & 0 & 3 \\ -2 & 2 & t \end{vmatrix} = 0,$$

从而有 $t = 3$. (3 分)

四边形 $ABCD$ 的面积为

$$\begin{aligned} S_{ABCD} &= S_{\triangle ABD} + S_{\triangle ACD} = \frac{1}{2} |\overrightarrow{AB} \times \overrightarrow{AD}| + \frac{1}{2} |\overrightarrow{AC} \times \overrightarrow{AD}| \\ &= \frac{1}{2} \left\| \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 2 & 0 \\ -2 & 2 & 3 \end{vmatrix} \right\| + \frac{1}{2} \left\| \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -1 & 0 & 3 \\ -2 & 2 & 3 \end{vmatrix} \right\| = \frac{1}{2} |(6, 3, 2)| + \frac{1}{2} |(-6, -3, -2)| = \frac{7}{2} + \frac{7}{2} = 7. \quad (6 \text{ 分}) \end{aligned}$$

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12. (6分) 【解】 易知

$$\frac{\partial z}{\partial x} = f' + 2xyg'_1 + 3g'_2, \quad (2 \text{ 分})$$

于是

$$\begin{aligned} \frac{\partial^2 z}{\partial x \partial y} &= \frac{\partial}{\partial y} (f' + 2xyg'_1 + 3g'_2) \\ &= -f'' + 2xg'_1 + 2xy(x^2g''_{11} - 4g''_{12}) + 3(x^2g''_{21} - 4g''_{22}) \\ &= -f'' + 2xg'_1 + 2x^3yg''_{11} + (3x^2 - 8xy)g''_{12} - 12g''_{22}. \end{aligned} \quad (6 \text{ 分})$$

13. (6分) 【解】 令 $x = \rho \cos \theta$, $y = \rho \sin \theta$, 则积分区域为

$$D: -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}, \quad 0 \leq \rho \leq \cos \theta. \quad (2 \text{ 分})$$

利用对称性, 有

$$\begin{aligned} \iint_D (x+y)^2 d\sigma &= \iint_D (x^2 + y^2) d\sigma = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} d\theta \int_0^{\cos \theta} \rho^2 \cdot \rho d\rho \\ &= \frac{1}{4} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos^4 \theta d\theta = \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos^4 \theta d\theta = \frac{1}{2} \cdot \frac{3 \cdot 1}{4 \cdot 2} \cdot \frac{\pi}{2} = \frac{3\pi}{32}. \end{aligned} \quad (6 \text{ 分})$$

14. (6分) 【解】 该物体的质量可表示为三重积分

$$M = \iiint_{\Omega} f(x, y, z) dV = \iiint_{\Omega} \sqrt{x^2 + y^2} dV. \quad (2 \text{ 分})$$

令 $x = r \sin \varphi \cos \theta$, $y = r \sin \varphi \sin \theta$, $z = r \cos \varphi$, 则

$$\begin{aligned} M &= \int_0^{2\pi} d\theta \int_0^{\pi} d\varphi \int_1^{\sqrt{2}} r \sin \varphi \cdot r^2 \sin \varphi dr \\ &= 2\pi \cdot \frac{3}{4} \cdot \int_0^{\pi} \sin^2 \varphi d\varphi = 2\pi \cdot \frac{3}{4} \cdot 2 \cdot \frac{1}{2} \cdot \frac{\pi}{2} = \frac{3\pi^2}{4}. \end{aligned} \quad (6 \text{ 分})$$

15. (6分) 【解】 记 $P(x, y) = x - 2y + 1$, $Q(x, y) = ax + y^2 + 3$, 则当

$$\frac{\partial Q}{\partial x} = a = \frac{\partial P}{\partial y} = -2$$

即 $a = -2$ 时, 存在可微函数 $u(x, y)$, 使 $du = (x - 2y + 1)dx + (ax + y^2 + 3)dy$. (3分)

易知

$$\begin{aligned} &(x - 2y + 1)dx + (-2x + y^2 + 3)dy \\ &= (x + 1)dx + (y^2 + 3)dy - (2ydx + 2xdy) \end{aligned}$$

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$$= d\left(\frac{1}{2}x^2 + \frac{1}{3}y^3 - 2xy + x + 3y\right) = d\left(\frac{1}{2}x^2 + \frac{1}{3}y^3 - 2xy + x + 3y\right),$$

故 $u(x, y) = \frac{1}{2}x^2 + \frac{1}{3}y^3 - 2xy + x + 3y + C$. 若 $u(0, 0) = 0$, 则所求 $u(x, y)$ 的表达式为

$$u(x, y) = \frac{1}{2}x^2 + \frac{1}{3}y^3 - 2xy + x + 3y. \quad (6 \text{ 分})$$

【另解】

$$\begin{aligned} u(x, y) &= \int_{(0,0)}^{(x,y)} (x-2y+1)dx + (-2x+y^2+3)dy + C \\ &= \int_0^x (x+1)dx + \int_0^y (-2x+y^2+3)dy + C \\ &= \frac{1}{2}x^2 + x - 2xy + \frac{1}{3}y^3 + 3y + C. \end{aligned}$$

若 $u(0, 0) = 0$, 则所求 $u(x, y)$ 的表达式为

$$u(x, y) = \frac{1}{2}x^2 + x - 2xy + \frac{1}{3}y^3 + 3y.$$

16. (6 分) 【解】记 $u_n(x) = \frac{n}{2n+1}x^{2n+1}$, 由于

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}(x)}{u_n(x)} \right| = \lim_{n \rightarrow \infty} \left| \frac{(n+1)x^{2n+3}}{2n+3} \cdot \frac{2n+1}{nx^{2n+1}} \right| = x^2,$$

由比值判别法知, 当 $x^2 < 1$, 即 $|x| < 1$ 时, $\sum_{n=0}^{\infty} \frac{n}{2n+1}x^{2n+1}$ 收敛; 当 $x^2 > 1$, 即 $|x| > 1$ 时,

$\sum_{n=0}^{\infty} \frac{n}{2n+1}x^{2n+1}$ 发散. 当 $x = \pm 1$ 时, 级数 $\pm \sum_{n=0}^{\infty} \frac{n}{2n+1}$ 发散, 因此幂级数 $\sum_{n=0}^{\infty} \frac{n}{2n+1}x^{2n+1}$ 的收

敛域为 $(-1, 1)$. (2 分)

记 $S(x) = \sum_{n=0}^{\infty} \frac{n}{2n+1}x^{2n+1}$, $-1 < x < 1$, 则

$$S(x) = \sum_{n=0}^{\infty} \frac{n}{2n+1}x^{2n+1} = \frac{1}{2} \sum_{n=0}^{\infty} x^{2n+1} - \frac{1}{2} \sum_{n=0}^{\infty} \frac{1}{2n+1}x^{2n+1}.$$

又记 $S_1(x) = \sum_{n=0}^{\infty} \frac{1}{2n+1}x^{2n+1}$, 对其逐项求导, 得

$$S_1'(x) = \sum_{n=0}^{\infty} x^{2n} = \frac{1}{1-x^2},$$

又 $S_1(0) = 0$, 将上式积分得

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$$S_1(x) - S_1(0) = \int_0^x S_1'(x) dx = \int_0^x \frac{1}{1-x^2} dx = \frac{1}{2} \int_0^x \left(\frac{1}{1-x} + \frac{1}{1+x} \right) dx = \frac{1}{2} \ln \frac{1+x}{1-x},$$

即

$$S_1(x) = \frac{1}{2} \ln \frac{1+x}{1-x}. \quad (5 \text{ 分})$$

$$\text{又 } \sum_{n=0}^{\infty} x^{2n+1} = x \sum_{n=0}^{\infty} x^{2n} = \frac{x}{1-x^2}, \text{ 故}$$

$$S(x) = \sum_{n=0}^{\infty} \frac{n}{2n+1} x^{2n+1} = \frac{1}{2} \frac{x}{1-x^2} - \frac{1}{4} \ln \frac{1+x}{1-x}. \quad (6 \text{ 分})$$

17. (8 分) 【解】方程组两边对 x 求导, 得

$$\begin{cases} 1 + \frac{dy}{dx} + \frac{dz}{dx} = 0, \\ 3x^2 + 3y^2 \frac{dy}{dx} - 3z^2 \frac{dz}{dx} = 0, \end{cases}$$

在点 $P(1,1,-2)$ 处, 化简上述方程组, 得

$$\begin{cases} \frac{dy}{dx} + \frac{dz}{dx} = -1, \\ \frac{dy}{dx} - 4 \frac{dz}{dx} = -1. \end{cases}$$

$$\text{从而在点 } P(1,1,-2) \text{ 处, } \frac{dy}{dx} = -1, \frac{dz}{dx} = 0. \quad (2 \text{ 分})$$

由于点 $P(1,1,-2)$ 处的切线方向与 y 轴正向夹角为锐角, 故 $\mathbf{u} = (-1, 1, 0)$, 对应的单位

$$\text{向量为 } \mathbf{e}_u = \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right). \quad (4 \text{ 分})$$

$$\text{由 } f'_x = \frac{2y}{1+2xy+yz}, \quad f'_y = \frac{2x+z}{1+2xy+yz}, \quad f'_z = \frac{y}{1+2xy+yz} \text{ 知}$$

$$f'_x(1,1,-2) = 2, \quad f'_y(1,1,-2) = 0, \quad f'_z(1,1,-2) = 1, \quad (6 \text{ 分})$$

于是

$$\begin{aligned} D_u f(1,1,-2) &= (f'_x, f'_y, f'_z) \Big|_{(1,1,-2)} \cdot \mathbf{e}_u \\ &= (2, 0, 1) \cdot \left(-\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, 0 \right) = -\sqrt{2}. \end{aligned} \quad (8 \text{ 分})$$

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18. (8分)【证明】首先求点 P 的坐标. 由于曲面 $S: F(x, y, z) = 0$ 上动点 (x, y, z) 到原点的距离为

$$d = \sqrt{x^2 + y^2 + z^2},$$

故问题转化为求目标函数 $x^2 + y^2 + z^2$ 在约束条件 $F(x, y, z) = 0$ 下的条件极小值点.

作拉格朗日函数

$$L(x, y, z, \lambda) = x^2 + y^2 + z^2 + \lambda F(x, y, z), \quad (2 \text{ 分})$$

令 $\nabla L = \mathbf{0}$, 得

$$\begin{cases} L'_x = 2x + \lambda F'_x(x, y, z) = 0, \\ L'_y = 2y + \lambda F'_y(x, y, z) = 0, \\ L'_z = 2z + \lambda F'_z(x, y, z) = 0, \\ L'_\lambda = F(x, y, z) = 0. \end{cases}$$

设 $P(x_0, y_0, z_0)$ 为曲面 $S: F(x, y, z) = 0$ 上离原点最近的点, 则 (x_0, y_0, z_0) 一定满足上述驻点方程组, 从而

$$F'_x(x_0, y_0, z_0) = -\frac{2}{\lambda} x_0, F'_y(x_0, y_0, z_0) = -\frac{2}{\lambda} y_0, F'_z(x_0, y_0, z_0) = -\frac{2}{\lambda} z_0. \quad (5 \text{ 分})$$

易知点 $P(x_0, y_0, z_0)$ 处的法线方程为

$$\frac{x - x_0}{F'_x(x_0, y_0, z_0)} = \frac{y - y_0}{F'_y(x_0, y_0, z_0)} = \frac{z - z_0}{F'_z(x_0, y_0, z_0)},$$

即

$$\frac{x - x_0}{x_0} = \frac{y - y_0}{y_0} = \frac{z - z_0}{z_0},$$

显然该法线方程经过原点. (8 分)

19. (8分)【解】取 Γ 所围曲面为 $\Sigma: z = x, \frac{x^2}{2} + \frac{y^2}{4} \leq 1$, 其法线方向指向上侧. 由斯托克斯

公式有

$$\begin{aligned} & \oint_{\Gamma} (x + y - y^2) dx + (z + x - x^2) dz \\ &= \iint_{\Sigma} \begin{vmatrix} dy dz & dz dx & dx dy \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x + y - y^2 & 0 & z + x - x^2 \end{vmatrix} = \iint_{\Sigma} (2x - 1) dz dx + (2y - 1) dx dy. \quad (3 \text{ 分}) \end{aligned}$$

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易知曲面 $\Sigma: z = x, \frac{x^2}{2} + \frac{y^2}{4} \leq 1$ 的法向量为 $\mathbf{n} = (1, 0, -1)$ ，故

$$\iint_{\Sigma} (2x-1) dz dx = \iint_{\Sigma} (2x-1) \cos \beta dS = 0, \quad (5 \text{ 分})$$

从而

$$\begin{aligned} \oint_{\Gamma} (x+y-y^2) dx + (z+x-x^2) dz &= \iint_{\Sigma} (2y-1) dx dy \\ &= \iint_{\frac{x^2}{2} + \frac{y^2}{4} \leq 1} (2y-1) dx dy = 0 - \iint_{\frac{x^2}{2} + \frac{y^2}{4} \leq 1} dx dy = -\pi \cdot \sqrt{2} \cdot 2 = -2\sqrt{2}\pi. \end{aligned}$$

其中由对称性知 $\iint_{\frac{x^2}{2} + \frac{y^2}{4} \leq 1} 2y dx dy = 0$. (8 分)

【另解】 曲线 Γ 的参数方程为 $x = \sqrt{2} \cos t, y = 2 \sin t, z = \sqrt{2} \cos t, t: 0 \rightarrow 2\pi$, (2 分)

则 $\oint_{\Gamma} (x+y-y^2) dx + (z+x-x^2) dz$

$$= \int_0^{2\pi} \left[(\sqrt{2} \cos t + 2 \sin t - 4 \sin^2 t) \cdot (-\sqrt{2} \sin t) + (2\sqrt{2} \cos t - 2 \cos^2 t) \cdot (-\sqrt{2} \sin t) \right] dt$$

(5 分)

$$\begin{aligned} &= \int_0^{2\pi} (-3 \sin 2t - 2\sqrt{2} \sin^2 t + 4\sqrt{2} \sin^3 t + 2\sqrt{2} \sin t \cos^2 t) dt \\ &= \frac{3}{2} \cos 2t \Big|_0^{2\pi} - 2\sqrt{2} \left(\frac{1}{2} t - \frac{1}{4} \sin 2t \right) \Big|_0^{2\pi} + 4\sqrt{2} \left(\frac{1}{3} \cos^3 t - \cos t \right) \Big|_0^{2\pi} - \frac{2\sqrt{2}}{3} \cos^3 t \Big|_0^{2\pi} \\ &= -2\sqrt{2} \pi. \end{aligned} \quad (8 \text{ 分})$$

20. (10 分) 【解】 所求流量为

$$\Phi = \iint_{\Sigma} (x^2 - xy) dy dz + (y^2 - yz) dz dx - (z^2 - zx) dx dy,$$

其中旋转曲面 $\Sigma: y = 1 + z^2 + x^2 (1 \leq y \leq 3)$ 的法向指向外侧. (2 分)

令 $P(x, y, z) = x^2 - xy, Q(x, y, z) = y^2 - yz, R(x, y, z) = z^2 - zx$, 则

$$\frac{\partial P}{\partial x} + \frac{\partial Q}{\partial y} + \frac{\partial R}{\partial z} = 2x - y + 2y - z + 2z - x = x + y + z.$$

取圆片 $\Sigma_1: \begin{cases} x^2 + z^2 \leq 2, \\ y = 3, \end{cases}$ 其法向与 y 轴正向相同. 设 Σ, Σ_1 所围成的立体为 Ω , 则由高斯公

式, 有

$$\oint_{\Sigma + \Sigma_1} (x^2 - xy) dy dz + (y^2 - yz) dz dx - (z^2 - zx) dx dy$$

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$$\begin{aligned}
&= \iiint_{\Omega} (x+y+z) dV = \iiint_{\Omega} y dV = \int_1^3 y dy \iint_{z^2+x^2 \leq y-1} dz dx \\
&= \pi \int_1^3 y(y-1) dy = \frac{14\pi}{3}.
\end{aligned} \quad (7 \text{ 分})$$

又

$$\begin{aligned}
&\iint_{\Sigma_1} (x^2 - xy) dy dz + (y^2 - yz) dz dx - (z^2 - zx) dx dy \\
&= \iint_{\Sigma_1} (y^2 - yz) dz dx = \iint_{\Sigma_1} (9 - 3z) dz dx = \iint_{z^2+x^2 \leq 2} (9 - 3z) dz dx \\
&= \iint_{z^2+x^2 \leq 2} 9 dz dx = 9 \cdot \pi \cdot 2 = 18\pi,
\end{aligned} \quad (9 \text{ 分})$$

故所求流量为

$$\Phi = \frac{14\pi}{3} - 18\pi = -\frac{40\pi}{3}. \quad (10 \text{ 分})$$

21. (10 分) 【解】 (1) 由题设可知, 在 t 时刻总有 $N = S(t) + I(t)$. 依题意, 有

$$\frac{dI}{dt} = \lambda S(t)I(t) - \mu I(t) = \lambda(N - I(t))I(t) - \mu I(t),$$

即求解下列初值问题

$$\begin{cases} \frac{dI}{dt} = (\lambda N - \mu - \lambda I)I, \\ I(0) = I_0. \end{cases} \quad (3 \text{ 分})$$

若 $\lambda N - \mu = 0$, 则原方程为 $\frac{dI}{dt} = -\lambda I^2$, 分离变量得

$$\frac{dI}{-I^2} = \lambda dt,$$

两边积分可得 $\frac{1}{I} = \lambda t + C$, 即 $I = \frac{1}{\lambda t + C}$. 代入初值条件 $I(0) = I_0$, 可得 $C = \frac{1}{I_0}$, 从而

$$I(t) = \frac{I_0}{1 + \lambda I_0 t}. \quad (5 \text{ 分})$$

若 $\lambda N - \mu \neq 0$, 则对原方程分离变量并整理得

$$dt = \frac{dI}{(\lambda N - \mu - \lambda I)I} = \frac{1}{\lambda N - \mu} \left(\frac{\lambda}{\lambda N - \mu - \lambda I} + \frac{1}{I} \right) dI,$$

两边积分 $\int \left(\frac{\lambda}{\lambda N - \mu - \lambda I} + \frac{1}{I} \right) dI = \int (\lambda N - \mu) dt$, 有

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$$\ln|I| - \ln|\lambda N - \mu - \lambda I| = (\lambda N - \mu)t + \ln|C|,$$

故

$$\frac{I}{\lambda N - \mu - \lambda I} = Ce^{(\lambda N - \mu)t},$$

解得 $I = \frac{(\lambda N - \mu)Ce^{(\lambda N - \mu)t}}{1 + \lambda Ce^{(\lambda N - \mu)t}}$. 代入初值条件 $I(0) = I_0$, 可得 $C = \frac{I_0}{\lambda N - \mu - \lambda I_0}$, 从而

$$I(t) = \frac{(\lambda N - \mu)I_0 e^{(\lambda N - \mu)t}}{\lambda I_0 e^{(\lambda N - \mu)t} + (\lambda N - \mu - \lambda I_0)} = \frac{(\lambda N - \mu)I_0}{\lambda I_0 + (\lambda N - \mu - \lambda I_0)e^{-(\lambda N - \mu)t}}.$$

综上所述, 有

$$I(t) = \begin{cases} \frac{(\lambda N - \mu)I_0}{(\lambda N - \mu - \lambda I_0)e^{-(\lambda N - \mu)t} + \lambda I_0}, & \frac{\mu}{\lambda} \neq N, \\ \frac{I_0}{1 + \lambda I_0 t}, & \frac{\mu}{\lambda} = N. \end{cases} \quad (7 \text{ 分})$$

(2) 由 (1) 可知, 当 $\frac{\mu}{\lambda} \geq N$ 时, $\lim_{t \rightarrow +\infty} I(t) = 0$, 即感染人数清零.

有效的防疫建议: 一方面良好的医疗水平和条件提高治愈率, 另一方面健康人采用戴口罩等防范措施, 病人采取严格隔离措施减少接触率. (10 分)

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